Algorithm 1: Baseline label-setting algorithm for maximum country coverage

Input:

- V. Set of nodes
- E_s. Set of scheduled edges (v, u, dep, arr)
- E_t. Set of transfer edges (v, w, Δ)
- country(v). Country associated with node v

Output:

- P. Sequence of labels representing an optimal route within 24 hours

Initialize Labels$(v) \leftarrow \emptyset$ for all $v \in V$
Initialize queue $Q \leftarrow \emptyset$
Initialize best_label $\leftarrow$ null
// *Initialize labels at departures*
for all $v \in V$ **do**
  for all scheduled departure from v at time dep with arrival arr **do**
    $L \leftarrow (v, dep, \{\text{country}(v)\}, \text{pred} = \text{null}, \text{start\_time} = dep)$
    Labels$(v) \leftarrow$ Labels$(v) \cup \{L\}$
    $Q \leftarrow Q \cup \{L\}$
// *Main loop*
while $Q \neq \emptyset$ **do**
  extract $L = (v, t, C, \text{pred}, \text{start\_time})$ from Q
  // *Only consider labels within 24-hour journey span*
  if $t - L.\text{start\_time} > 24:00$ **then**
    continue
  if best_label $=$ null $\vee\ |C| > |\text{best\_label}.C|$ **then**
    best_label $\leftarrow L$
  // *Extend via scheduled edges*
  for all $(v, u, dep, arr) \in E_s$ **do**
    if $dep \geq t \wedge dep - L.\text{start\_time} \leq 24:00$ **then**
      $t_{\text{new}} \leftarrow arr$
      if $t_{\text{new}} - L.\text{start\_time} > 24:00$ **then**
        continue
      $C' \leftarrow C \cup \{\text{country}(u)\}$
      $L' \leftarrow (u, t_{\text{new}}, C', \text{pred} = L, \text{start\_time} = L.\text{start\_time})$

      if $\nexists L'' \in$ Labels(u) such that
        $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C' \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C'))$ **then**
        remove all $L'' \in$ Labels(u) such that
          $(t_{\text{new}} \leq L''.t \wedge C' \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C' \neq L''.C))$
        Labels$(u) \leftarrow$ Labels$(u) \cup \{L'\}$
        $Q \leftarrow Q \cup \{L'\}$
  // *Extend via transfers*
  for all $(v, w, \Delta) \in E_t$ **do**
    $t_{\text{new}} \leftarrow t + \Delta$
    if $t_{\text{new}} - L.\text{start\_time} > 24:00$ **then**
      continue
    $L' \leftarrow (w, t_{\text{new}}, C, \text{pred} = L, \text{start\_time} = L.\text{start\_time})$

if $\nexists L'' \in \text{Labels}(w)$ such that
$(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C))$ **then**
remove all $L'' \in \text{Labels}(w)$ such that
$(t_{\text{new}} \leq L''.t \wedge C \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C \neq L''.C))$
$\text{Labels}(w) \leftarrow \text{Labels}(w) \cup \{L'\}$
$Q \leftarrow Q \cup \{L'\}$
// *Reconstruct path*
$P \leftarrow$ empty sequence
$L \leftarrow$ best_label
while $L \neq$ null **do**
prepend L to P
$L \leftarrow L$.pred
return P

This is a very naive approach which is totally unfeasible, since for over 200 countries the number of states grows combinatorially, resulting in extremely large running times. Continuous transfers do not contribute to that task either.

On the other hand, this does not handle the days of the events. Since we don't know when our algorithm will choose to start the journey, we'll provably need data spanning over 48 hours, and thus having events over different days. Hence, the substractions made here would make no sense if substracting hours from dfferent ways. This should be considered when implementing the final algorithm in code.